# Ilias Papanikolaou

PhD Student in Computer Science, Northwestern University

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## Education

**Northwestern University** — Evanston, IL, USA
*Ph.D. in Theoretical Computer Science* — *Sep. 2023 – Present (expected 2028)*
*Advisor: Prof. Konstantin Makarychev*

**National Technical University of Athens** — Athens, Greece
*Integrated Master (MEng) in Electrical and Computer Engineering, GPA: 8.93/10* — *Oct. 2017 – Apr. 2023*
*Diploma thesis: "Explainable Clustering Under Perturbation Stability"* — *Sep. 2022 – Apr. 2023*
*Advisor: Prof. Dimitris Fotakis*

## Publications

*Authors contributed equally and are listed in alphabetical order, as is common in theoretical computer science.*

**A Simple Proof of the Exact $(H_{k-1}+1)$ Competitive Ratio for Explainable $\ell_1$ Clustering**
Konstantin Makarychev, Yury Makarychev, **Ilias Papanikolaou**, Liren Shan
*To appear in the SIAM Symposium on Simplicity in Algorithms (SOSA), 2027*

**Near-optimal Explainable $k$-means Clustering under $\ell_p$ Norm**
Xinyuan Cao, Konstantin Makarychev, **Ilias Papanikolaou**, Liren Shan
*To appear in the Conference on Neural Information Processing Systems (NeurIPS), 2026*

**Dynamic Algorithm for Explainable $k$-medians Clustering under the $\ell_p$ Norm** [arXiv]
Konstantin Makarychev, **Ilias Papanikolaou**, Liren Shan
*Conference on Neural Information Processing Systems (NeurIPS), 2025.* **Spotlight Presentation**

## Teaching Experience

**Northwestern University**, Teaching Assistant

- CS436: Graduate Algorithms — Fall 2026
- CS336: Design and Analysis of Algorithms — Winter 2026
- CS212: Discrete Math — Winter 2025

**National Technical University of Athens**

- Algorithms and Complexity, Teaching Assistant — 2022
- Computer Programming, Lab Assistant — 2022

## Academic Service

**Reviewer**: International Conference on Learning Representations (ICLR) 2027

## Skills and Languages

**Programming**: C/C++, Python (PyTorch), Java
**LLM Experience** (CS461 - Deep Learning for Natural Language Processing): implementation of neural language models; fine-tuning of BERT and GPT-style models for question answering and text generation, including LoRA
**Languages**: Greek (native), English (full professional proficiency), German (elementary)